

Lecture 4: Newton's Method: Derivation

Math 6610

C. Mou

Department of Mathematics and Statistics
Utah State University

Fall 2026

Today's Goals

- Recap the LU toolkit we now rely on for linear solves

Today's Goals

- Recap the LU toolkit we now rely on for linear solves
- Formulate root-finding $f(x) = 0$ for n nonlinear equations

Today's Goals

- Recap the LU toolkit we now rely on for linear solves
- Formulate root-finding $f(x) = 0$ for n nonlinear equations
- Derive Newton's method for $n = 1$ from linearization

Today's Goals

- Recap the LU toolkit we now rely on for linear solves
- Formulate root-finding $f(x) = 0$ for n nonlinear equations
- Derive Newton's method for $n = 1$ from linearization
- See with $f = \tan^{-1}$ that Newton can cycle or diverge

LU Factorization Recap

$$PA = LU$$
$$L = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ l_{21} & 1 & & \vdots \\ \vdots & & \ddots & 0 \\ l_{n1} & \cdots & l_{n,n-1} & 1 \end{bmatrix}, \quad |l_{jk}| \leq 1$$

LU factorization with partial pivoting is stable in practice.

Newton's Method

Newton's method is not a finite method, meaning that you would have to run it infinitely many times in order for it to arrive at the answer.

Remark (Newton's Method Overview). Given: $f : D \rightarrow \mathbb{R}^n$, $D \subset \mathbb{R}^n$

$$f = \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_n \end{bmatrix}, \quad f_j : D \rightarrow \mathbb{R}, \quad j = 1, 2, \dots, n$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in D \subset \mathbb{R}^n, \quad x_j \in \mathbb{R}, \quad j = 1, 2, \dots, n$$

Goal: Find $x \in D$ such that

$$f(x) = 0.$$

Thus, we have n nonlinear equations for n unknowns.

Newton's Method

Example (Newton's Method Example 1: Possible Scenarios). $D = \mathbb{R}^n$,
 $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$,

$$f(x) = b - Ax, \quad b \in \mathbb{R}^n, A \in \mathbb{R}^{n \times n}$$

$$f(x) = 0 \Leftrightarrow Ax = b$$

There are 3 possibilities:

- ❶ 1 solution
- ❷ no solution
- ❸ infinitely many solutions

Example (Newton's Method Example 2). $f = f_\alpha: \mathbb{R} \rightarrow \mathbb{R}$, $f_\alpha(x) = x^2 + \alpha$,
 $\alpha \in \mathbb{R}$ is a parameter. This is an upward facing parabola with minimum value α .

- ❶ $f(x)$ has no solution if $\alpha > 0$
- ❷ $f(x)$ has one solution if $\alpha = 0$ ($x = 0$)
- ❸ $f(x)$ has two solutions if $\alpha < 0$ ($x = \pm \sqrt{-\alpha}$)

Newton's Method

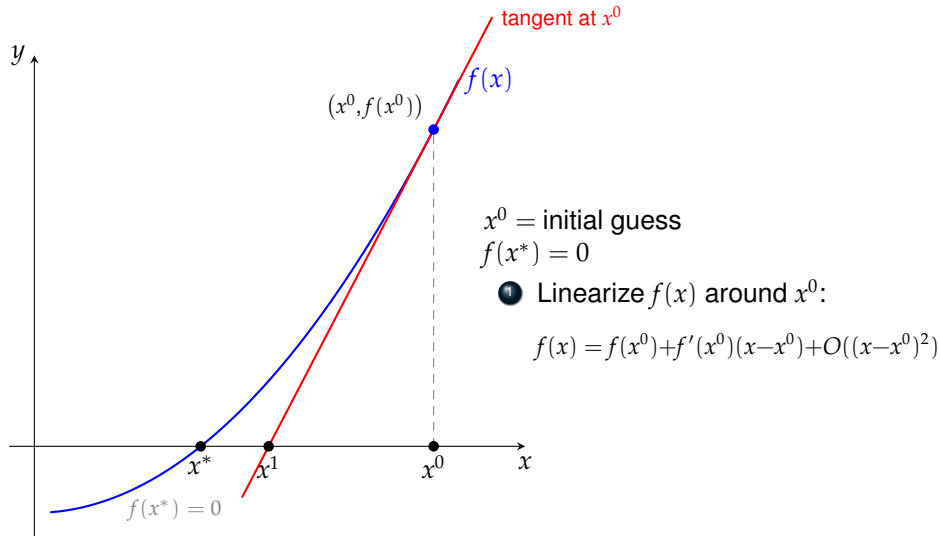
Example (Newton's Method Example 3). $n = 1$, $D = \mathbb{R}^+ = \{x \in \mathbb{R} \mid x > 0\}$, $f(x) = \sin \frac{1}{x}$. Thus, f oscillates faster as $x \rightarrow 0$.

$$f(x) = 0 \Leftrightarrow \frac{1}{x} = j\pi, \quad j = 1, 2, \dots$$

$$x = \frac{1}{j\pi}$$

Newton's Method

Remark (Newton's Method Algorithm for $n = 1$).



Newton's Method

- 2 Get new approximation x^1 by setting $t(x^1) = 0$ and solving for x^1 :

$$x^1 = x^0 - \frac{f(x^0)}{f'(x^0)}$$

(provided $f'(x) \neq 0$).

- 3 Repeat with x^0 replaced by x^1 . Iterative process:

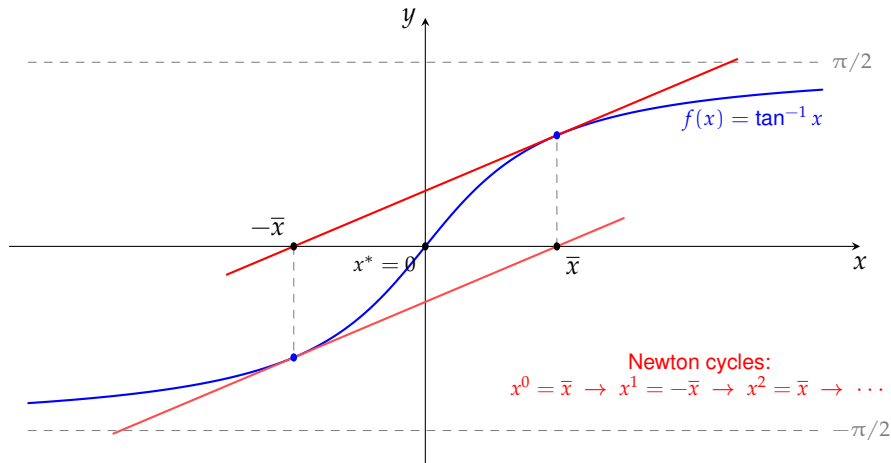
$$x^{k+1} = x^k - \frac{f(x^k)}{f'(x^k)}, \quad k = 0, 1, 2, \dots$$

Remark (*Questions about Newton's Method*).

- 1 Can we guarantee that $\lim_{k \rightarrow \infty} x^k = x^*$?
- 2 If yes, what is the speed of convergence?

Newton's Method

Example (Failure of Newton's Method).



$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = \tan^{-1} x, \quad f'(x) = \frac{1}{1+x^2}.$$

f has a single zero: $x^* = 0$. We apply Newton's method. Choose $x^0 \in \mathbb{R}$.

$$x^{k+1} = x^k - (1 + (x^k)^2) \tan^{-1}(x^k), \quad k = 0, 1, 2, \dots$$

Newton's Method

Convergence?

Let $\bar{x}$ be the unique solution of

$$\begin{aligned}-\bar{x} &= \bar{x} - (1 + \bar{x}^2) \tan^{-1} \bar{x}, & \bar{x} > 0 \\ \bar{x} &= 1.39174 \dots\end{aligned}$$

where $\bar{x}$ is computed by using Newton's Method for

$$g(x) = 2x - (1 + x^2) \tan^{-1} x.$$

Thus, if $x^0 = \bar{x}$, then $x^k = (-1)^k \bar{x}$, $k = 0, 1, \dots$

Similarly, if $x^0 = -\bar{x}$, then $x^{k+1} = (-1)^{k+1} \bar{x}$, $k = 0, 1, \dots$

If $|x^0| < \bar{x}$, then $\lim_{k \rightarrow \infty} x^k = 0 = x^*$

If $|x^0| > \bar{x}$, then $\lim_{k \rightarrow \infty} x^k = \infty$

Filling in the Details: The Linearization Step

Everything comes from Taylor's theorem at the current guess x^0 :

$$f(x) = \underbrace{f(x^0) + f'(x^0)(x - x^0)}_{=:t(x), \text{ tangent line}} + \underbrace{O((x - x^0)^2)}_{\text{ignored}}.$$

We cannot solve $f(x) = 0$, but we *can* solve $t(x) = 0$:

$$0 = f(x^0) + f'(x^0)(x^1 - x^0) \implies x^1 = x^0 - \frac{f(x^0)}{f'(x^0)}.$$

Geometrically: slide down the tangent line to the x -axis.

The neglected term is $O((x - x^0)^2)$: if the current error is small, the *new* error is roughly its square. This is the germ of quadratic convergence – doubling of correct digits – which we will prove in Lectures 6–7.

Filling in the Details: The Linearization Step

A worked example to keep in mind: computing $\sqrt{a}$ means solving $f(x) = x^2 - a = 0$. Newton gives

$$x^{k+1} = x^k - \frac{(x^k)^2 - a}{2x^k} = \frac{1}{2} \left(x^k + \frac{a}{x^k} \right).$$

For $a = 2$, $x^0 = 1$:

$$x^1 = 1.5, \quad x^2 = 1.41\bar{6}, \quad x^3 = 1.4142157\dots, \quad x^4 = 1.4142136\dots$$

Correct digits: 1, 3, 6, 12, $\dots$ – they double each step. This is what quadratic convergence looks like in practice.

Understanding the $\tan^{-1}$ Failure

The iteration is $x^{k+1} = g(x^k)$ with $g(x) = x - (1 + x^2) \tan^{-1} x$. Since g is odd, a 2-cycle $x \rightarrow -x \rightarrow x$ occurs exactly when

$$g(\bar{x}) = -\bar{x} \iff 2\bar{x} = (1 + \bar{x}^2) \tan^{-1} \bar{x},$$

which has the unique positive solution $\bar{x} = 1.39174\dots$

- $|x^0| < \bar{x}$: each step overshoots but lands *closer*; convergence to $x^* = 0$.
- $|x^0| = \bar{x}$: perpetual ping-pong $\bar{x} \rightarrow -\bar{x} \rightarrow \bar{x} \rightarrow \dots$
- $|x^0| > \bar{x}$: overshoot grows; $|x^k| \rightarrow \infty$.

Takeaways: (1) Newton's convergence is *local* – guarantees require x^0 close enough to x^* ; (2) flat functions ($f' \rightarrow 0$ far out) make the Newton step $-f/f'$ huge. Both issues motivate the convergence theorem (Lec. 6) and damping (Lec. 7–8).

Looking Ahead

- **Next lecture:** Newton's Method: Local Convergence